FIRMWARE OVER-THE-AIR AT SCALE

Updating firmware is easy. But what happens when millions of connected devices need updates at once? Explore how FOTA handles them at scale—without costly service visits.



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As connected devices proliferate across industries, organisations need firmware-over-the-air (FOTA) systems that can manage thousands or millions of devices from a single platform. A scalable FOTA architecture lets organisations monitor firmware versions, hardware revisions, and deployment status, while remotely updating devices regardless of the software version currently installed.

FOTA is becoming increasingly important as product development cycles shorten. Companies are launching products faster, often in smaller batches, while customer requirements can change throughout the product lifecycle. Since not every real-world condition can be tested, software bugs can emerge after deployment. FOTA allows organisations to remotely deploy bug fixes, performance improvements, new functionality, and security updates, reducing the need for recalls and on-site maintenance.

FOTA is widely used in sectors with large-scale deployments. Telematics, fleet management, and mobility applications require remote firmware updates because devices are often distributed across large geographical areas. The same applies to EV charging infrastructure. More broadly, IoT devices using Wi-Fi, cellular, Bluetooth, or other communication technologies are increasingly expected to support secure over-the-air updates.

In addition to making maintenance easier, FOTA can extend the lifespan of connected devices. For instance, if a network operator ceases support for a specific technology, devices may become unusable unless they receive a firmware upgrade. Devices compatible with NB-IoT, for example, can be upgraded to support the LTE Cat-M1 network through a FOTA update when an operator phases out NB-IoT networks. This shows that FOTA is no longer simply a way to reduce maintenance costs, it can also